

Financial Mathematics 32000

Lecture 6

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Variance reduction techniques

Recall the radius of a $100(1 - p)\%$ confidence interval is

$$\Phi^{-1}\left(1 - \frac{p}{2}\right) \sqrt{\text{Var}(\hat{C}_M)}$$

Variance reduction techniques $\rightarrow$ estimators with lower Var than $\hat{C}_M$

- ▶ Introduce dependence across simulations:
 - ▶ Antithetic variates $\hat{C}_M^{\text{av}}$
- ▶ Simulate something other than the contract's payoff:
 - ▶ Control variates $\hat{C}_M^{\text{cv}}$
 - ▶ Conditional Monte Carlo $\hat{\sigma}_M^{\text{cmc}}$
- ▶ Modify the probability distribution:
 - ▶ Importance sampling $\hat{C}_M^{\text{ris}}$

(Increasing M reduces Var, but not described as “variance reduction”)

Variance reduction: Antithetic variates

Variance reduction: Control variates

Variance reduction: Importance sampling

Variance reduction: Conditional MC

Antithetic variates

Let Y be discounted payoff. So

$$C = \mathbb{E}Y.$$

Ordinary MC: $\hat{C}_M = \frac{1}{M}(Y_1 + \dots + Y_M)$, where each Y_m is IID as Y .

Example: Call on stock under GBM:

$$Y_m = e^{-rT}(S_0 e^{(r-\sigma^2/2)T + \sigma\sqrt{T}Z_m} - K)^+$$

where we generate the Z_m to be IID $N(0, 1)$ for $m = 1, \dots, M$.

Antithetic Variates: If you draw a realization from the q th percentile, then you should *also draw one from the $(100 - q)$ th percentile*.

Symmetry of normal $\Rightarrow$ for each realization of Z , rerun also with $-Z$.

Antithetic variates

In the same example, for each $m = 1, \dots, M$, let

$$\tilde{Z}_m := -Z_m$$

$$\tilde{Y}_m := e^{-rT} (S_0 e^{(r-\sigma^2/2)T + \sigma\sqrt{T}\tilde{Z}_m} - K)^+$$

$$Y_m^{\text{av}} := \frac{Y_m + \tilde{Y}_m}{2}$$

$$\hat{C}_M^{\text{av}} := \frac{1}{M} (Y_1^{\text{av}} + Y_2^{\text{av}} + \dots + Y_M^{\text{av}})$$

This is the antithetic-variate Monte Carlo estimate.

Its expectation is

$$\mathbb{E}\hat{C}_M^{\text{av}} = \frac{1}{M} \sum_m \mathbb{E}Y_m^{\text{av}} = \mathbb{E}\left(\frac{Y + \tilde{Y}}{2}\right) = \frac{\mathbb{E}Y + \mathbb{E}\tilde{Y}}{2} = \mathbb{E}Y = C$$

so $\hat{C}_M^{\text{av}}$ is an unbiased estimate of C .

Antithetic variates: variance analysis

Variance of AV estimate is

$$\begin{aligned}\text{Var}(\hat{C}_M^{\text{av}}) &= \text{Var}\left(\frac{1}{M} \sum_m Y_m^{\text{av}}\right) = \frac{1}{M^2} M \text{Var}(Y^{\text{av}}) \\ &= \frac{1}{M} \text{Var}\left(\frac{Y + \tilde{Y}}{2}\right) = \frac{1}{M} \frac{\text{Var}Y + 2\text{Cov}(Y, \tilde{Y}) + \text{Var}(\tilde{Y})}{4} \\ &= \frac{1}{M} \left(\frac{1}{2} \text{Var}Y + \frac{1}{2} \text{Cov}(Y, \tilde{Y})\right)\end{aligned}$$

Comparing to ordinary MC,

$$\text{Var}(\hat{C}_M) = \frac{1}{M} \text{Var}(Y).$$

and $\text{Cov}(Y, \tilde{Y}) \leq \sqrt{\text{Var}(Y)\text{Var}(\tilde{Y})} = \text{Var}(Y)$ so $\text{Var}(\hat{C}_M^{\text{av}}) \leq \text{Var}(\hat{C}_M)$.

But maybe this overstates the benefit of AV.

Antithetic variates: variance analysis

Maybe better to compare

$$\text{Var}(\hat{C}_M^{\text{av}}) = \frac{1}{M} \left(\frac{1}{2} \text{Var}Y + \frac{1}{2} \text{Cov}(Y, \tilde{Y}) \right)$$

against

$$\text{Var}(\hat{C}_{2M}) = \frac{1}{2M} \text{Var}(Y).$$

So $\text{Var}(\hat{C}_M^{\text{av}})$ is smaller iff $\text{Cov}(Y, \tilde{Y}) < 0$.

This often holds, but for some payoffs, it doesn't.

Antithetic variates: variance analysis

Because

$$\text{Var}(\hat{C}_M^{\text{av}}) = \frac{1}{M} \text{Var}(Y^{\text{av}}),$$

an asymptotic $100(1 - p)\%$ confidence interval has endpoints

$$\hat{C}_M^{\text{av}} \pm \Phi^{-1}\left(1 - \frac{p}{2}\right) \frac{\hat{\sigma}_M^{\text{av}}}{\sqrt{M}}$$

where

$$\hat{\sigma}_M^{\text{av}} := \sqrt{\frac{1}{M-1} \sum_m (Y_m^{\text{av}} - \hat{C}_M^{\text{av}})^2}.$$

Note that $\hat{\sigma}_M^{\text{av}}$ is the sample standard deviation of the *pair averages*, not of the individual realizations.

Dividing it by $\sqrt{M}$ gives the standard error $\hat{\sigma}_M^{\text{av}}/\sqrt{M}$, an estimate of the standard deviation of $\hat{C}_M^{\text{av}} = \frac{1}{M} \sum_m Y_m^{\text{av}}$.

Variance reduction: Antithetic variates

Variance reduction: Control variates

Variance reduction: Importance sampling

Variance reduction: Conditional MC

Control variates

A control variate Y^* is a random variable, related to Y , such that

$C^* := \mathbb{E}Y^*$ is known (e.g. via explicit formula). Examples:

- ▶ Let $dS_t = \sigma_t S_t dW_t$, where σ_t is stochastic. If Y is the discounted payoff of a call on S , we can let Y^* be the discounted payoff of a call on S^* where $dS_t^* = \sigma S_t^* dW_t$. So C^* is known (B-S formula). Here σ is some constant that approximates the σ_t process.
- ▶ If Y is discounted payoff of Asian-style call on GBM:

$$Y = e^{-rT} \left(\frac{1}{T} \int_0^T S_t dt - K \right)^+$$

Can let Y^* be the discounted *European* call payoff on the GBM.

Or let Y^* be the discounted payoff of a *geometric Asian* call:

$Y^* = e^{-rT} \left(e^{\frac{1}{T} \int_0^T \log S_t dt} - K \right)^+$. In either case, C^* has a formula.

Control variates

The control variate technique simulates the **difference** $Y - \beta Y^*$, where β is a constant. The difference has expectation

$$\mathbb{E}(Y - \beta Y^*) = C - \beta C^*$$

and variance $\text{Var}(Y - \beta Y^*)$ which is, we hope, small. Therefore

$$C = \mathbb{E}Y + \beta(C^* - \mathbb{E}Y^*).$$

Replacing expectations by sample averages, we define the control variate Monte Carlo estimate of C to be

$$\hat{C}_M^{\text{cv},\beta} := \bar{Y}_M + \beta(C^* - \bar{Y}_M^*) \quad \text{where} \quad \begin{aligned} \bar{Y}_M &:= \frac{1}{M}(Y_1 + \cdots + Y_M) \\ \bar{Y}_M^* &:= \frac{1}{M}(Y_1^* + \cdots + Y_M^*) \end{aligned}$$

and we use the *same* pseudo-random numbers generate Y_m and Y_m^* .

Control variates: variance analysis

Expectation:

$$\mathbb{E}\hat{C}_M^{\text{cv},\beta} := \mathbb{E}\hat{C}_M + \beta(C^* - C^*) = C$$

so $\hat{C}_M^{\text{cv},\beta}$ is an unbiased estimate of C .

Variance:

$$\begin{aligned}\text{Var}(\hat{C}_M^{\text{cv},\beta}) &= \text{Var}(\bar{Y}_M - \beta\bar{Y}_M^*) = \frac{1}{M} \text{Var}(Y - \beta Y^*) \\ &= \frac{1}{M} [\text{Var}(Y) - 2\beta \text{Cov}(Y, Y^*) + \beta^2 \text{Var}(Y^*)]\end{aligned}$$

Now choose β to minimize this. Obtain

$$\beta_{\text{optimal}} = \text{Cov}(Y, Y^*) / \text{Var}(Y^*)$$

$$\text{and } \text{Var}(\hat{C}_M^{\text{cv},\beta_{\text{optimal}}}) = \text{Var}(\hat{C}_M) \times (1 - \text{Corr}^2(Y, Y^*)) \leq \text{Var}(\hat{C}_M)$$

Control variates: optimizing the coefficient

Unfortunately the Cov and Var are usually not both known.

But we can use the *sample* covariance and variance. Let

$$\hat{\beta} := \frac{\sum_m (Y_m - \bar{Y}_M)(Y_m^* - \bar{Y}_M^*)}{\sum_m (Y_m^* - \bar{Y}_M^*)^2}$$

And define

$$\hat{C}_M^{\text{cv}, \hat{\beta}} := \bar{Y}_M + \hat{\beta}(C^* - \bar{Y}_M^*).$$

Note $\mathbb{E}(C^* - \bar{Y}_M^*) = 0$ does not imply $\mathbb{E}[\hat{\beta}(C^* - \bar{Y}_M^*)] = 0$, so this creates **bias** in the estimate:

$$\mathbb{E}\hat{C}_M^{\text{cv}, \hat{\beta}} = C + \mathbb{E}[\hat{\beta}(C^* - \bar{Y}_M^*)].$$

As $M \rightarrow \infty$ the bias $\rightarrow 0$.

Control variates: confidence intervals

To compute confidence intervals for fixed β , note that

$$\hat{C}_M^{\text{cv},\beta} = \frac{1}{M} \sum_m Y_m^{\text{cv},\beta}$$

where the independent simulations

$$Y_m^{\text{cv},\beta} := Y_m + \beta(C^* - Y_m^*)$$

have an estimated variance

$$\frac{1}{M-1} \sum_m (Y_m^{\text{cv},\beta} - \hat{C}_M^{\text{cv},\beta})^2.$$

Let $\hat{\sigma}_M^{\text{cv},\beta}$ be the square root of this estimated variance.

Control variates: confidence intervals

Note that

$\hat{\sigma}_M^{\text{cv},\beta}$ = estimated standard deviation of *one* of the $Y_m^{\text{cv},\beta}$

$\hat{\sigma}_M^{\text{cv},\beta}/\sqrt{M}$ = estimated st. dev. of $\hat{C}_M^{\text{cv},\beta}$, the *average* of the $Y_m^{\text{cv},\beta}$

Then an asymptotic $100(1-p)\%$ confidence interval has endpoints

$$\hat{C}_M^{\text{cv},\beta} \pm \Phi^{-1}\left(1 - \frac{p}{2}\right) \frac{\hat{\sigma}_M^{\text{cv},\beta}}{\sqrt{M}}.$$

If instead of β we use $\hat{\beta}$ estimated from those M simulations, the variance estimate is biased for finite M , but as $M \rightarrow \infty$ its bias and variance $\rightarrow 0$, so the same confidence interval calculation (with $\hat{\beta}$ in place of β) is asymptotically valid.

Variance reduction: Antithetic variates

Variance reduction: Control variates

Variance reduction: Importance sampling

Variance reduction: Conditional MC

Motivating example

Using Monte Carlo, a rare event with a big payoff is harder to price than a more frequent event with a smaller payoff.

- ▶ Consider an asset that pays 100 dollars with probability 2% (under some pricing measure), and pays zero otherwise.

Its price is 2 and the payoff variance is $10000 \times 0.02 - 2^2 = 196$

- ▶ Now consider an asset that pays 10 dollars with probability 20%, zero otherwise.

Its price is 2. But its payoff variance is only $100 \times 0.20 - 2^2 = 16$

So maybe if we *sample from a different distribution* and *modify the payoff* we can preserve the expectation, while decreasing the variance.

Importance sampling

Let X be a random vector in $\mathbb{R}^N$, with density function $f : \mathbb{R}^N \rightarrow \mathbb{R}$.

Let $h : \mathbb{R}^N \rightarrow \mathbb{R}$. Think of h as a discounted payoff function.

We want to estimate

$$C := \mathbb{E}h(X) = \int h(x)f(x)dx$$

Ordinary Monte Carlo:

$$\hat{C} := \frac{1}{M} \sum_{m=1}^M h(X_{[m]})$$

where $X_{[1]}, \dots, X_{[M]}$ are IID random draws from density f .

Can apply to X = some financial variable (such as an asset price), or
can apply to X = as the randomness that drives the dynamics (such
as a vector of Brownian increments).

Change of measure

Let g be any density on $\mathbb{R}^N$ that vanishes only where fh does, meaning $f(x)h(x) = 0$ if $g(x) = 0$. Let $A := \{x : f(x)h(x) \neq 0\}$. Then

$$C = \int_A h(x)f(x)dx = \int_A h(x)\frac{f(x)}{g(x)}g(x)dx = \mathbb{E}^*\left[h(X)\frac{f(X)}{g(X)}\right]$$

where $\mathbb{E}^*$ denotes expectation with respect to a new measure $\mathbb{P}^*$ under which X has density g .

The *importance sampling* Monte Carlo estimate of C is

$$\hat{C}^{\text{is}} := \frac{1}{M} \sum_{m=1}^M h(X_{[m]}) \frac{f(X_{[m]})}{g(X_{[m]})}$$

where $X_{[1]}, \dots, X_{[M]}$ are IID random draws from density g , which we call the *importance sampling* density.

Change of measure

So we have made two changes to the ordinary MC estimate

- ▶ Change the distribution from which X is sampled. Draw from density g instead of f .
- ▶ Change the payoff. Instead of $h(X)$, it becomes $h(X)$ times the *Radon-Nikodym derivative* $f(X)/g(X)$.

What about bias and variance?

- ▶ By construction, $\hat{C}^{\text{is}}$ is an unbiased estimator of C because

$$\mathbb{E}^* \hat{C}^{\text{is}} = \frac{1}{M} \sum_{m=1}^M \mathbb{E}^* \left[h(X_{[m]}) \frac{f(X_{[m]})}{g(X_{[m]})} \right] = C$$

- ▶ To minimize variance, try to choose g such that hf/g is constant on the set where $g \neq 0$. Try to “flatten the curve” (the payout h).

Variance

The importance-sampling MC estimate $\hat{C}^{\text{is}}$ has (wrt $\mathbb{P}^*$) variance

$$\frac{1}{M} \text{Var}^* \frac{h(X)f(X)}{g(X)} = \mathbb{E}^* \frac{h(X)^2 f(X)^2}{g(X)^2} - C^2 = \mathbb{E} \left[h(X)^2 \frac{f(X)}{g(X)} \right] - C^2$$

which can be bigger or smaller than the ordinary MC variance:

$$\frac{1}{M} \text{Var} h(X) = \mathbb{E}[h(X)^2] - C^2.$$

How to choose g to minimize $\text{Var}^* \hat{C}^{\text{is}}$? Answer: if $h \geq 0$ and

$$g(x) = \text{constant} \times h(x)f(x)$$

then

$$\text{Var}^* \hat{C}^{\text{is}} = \frac{1}{M} \text{Var}^* \frac{h(X)f(X)}{g(X)} = 0.$$

Variance reduction

- ▶ However, note that the unique constant such that

$$g(x) = \text{constant} \times h(x)f(x)$$

defines a legitimate density is $1 / \int h(x)f(x)dx = 1/C$.

If we knew the value of C , we wouldn't need Monte Carlo. So in practice, this recipe for choosing g cannot be followed exactly.

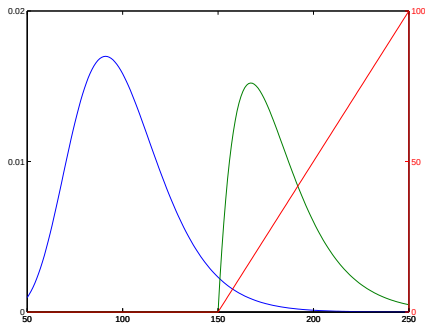
- ▶ What we *can* do is choose g to be a known density *approximately* proportional to the original density f times the payoff h .

Irrespective of whether this approximation is bad or good, we still have an unbiased estimate $\hat{C}^{\text{is}}$.

A good approximation can make the variance of $\hat{C}^{\text{is}}$ small.

Variance reduction

If $X = (X_1, \dots, X_N)$ are Brownian increments driving stock $S(X)$, where $S : \mathbb{R}^N \rightarrow \mathbb{R}$, then can apply this idea to $S(X)$ instead of X . In other words, if $h(x) = H(S(x))$, then variance is zero if $S(X)$ has IS-density = constant $\times$ H $\times$ original density of $S(X)$.



Density ratio computations

To compute $f(X)/g(X)$, usually you won't be specifying g . Rather you'd specify, for instance, a desired mean or drift, then compute g . Compute g by starting in X space.

Suppose, in particular, that the components of $X \in \mathbb{R}^N$ under the original measure are independent with densities $f_1, f_2, \dots, f_N$, and that under importance sampling measure (not necessarily optimal), they are independent with densities $g_1, g_2, \dots, g_N$. Then

$$\frac{f(x)}{g(x)} = \frac{f(x_1, \dots, x_N)}{g(x_1, \dots, x_N)} = \prod_{n=1}^N \frac{f_n(x_n)}{g_n(x_n)}$$

(Actually, independence is unnecessary, if you let f_n be the *conditional* density of X_n given $X_1, \dots, X_{n-1}$, and similarly for g_n .)

Example: Changing of drift of BM

Let $X_1, \dots, X_N$ be successive increments of BM W sampled at times $0, t_1, \dots, t_N = T$. Then density f_n is Normal($0, \Delta t$), where $\Delta t = T/N$.

$$f_n(x_n) = \frac{1}{\sqrt{2\pi\Delta t}} e^{-x_n^2/(2\Delta t)}$$

The g_n can be the density of any other distribution supported on $\mathbb{R}$.

In particular, if we take g_n to be the Normal($\lambda\Delta t, \Delta t$) density, then

$$g_n(x_n) = \frac{1}{\sqrt{2\pi\Delta t}} e^{-(x_n - \lambda\Delta t)^2/(2\Delta t)}$$

So for all $x = (x_1, \dots, x_n)$,

$$\frac{f_n(x_n)}{g_n(x_n)} = e^{\frac{1}{2\Delta t}(-x_n^2 + x_n^2 - 2x_n\lambda\Delta t + \lambda^2(\Delta t)^2)} = e^{-\lambda x_n + \frac{1}{2}\lambda^2\Delta t}$$

$$\frac{f(x)}{g(x)} = \prod_{n=1}^N \frac{f_n(x_n)}{g_n(x_n)} \implies \frac{f(X)}{g(X)} = e^{-\lambda \sum X_n + \frac{1}{2}\lambda^2 T} = e^{-\lambda W_T + \frac{1}{2}\lambda^2 T}$$

Example: Changing the drift of BM

Implementation

- ▶ If you want to change the drift of W to λ , then work under $\mathbb{P}^*$ such that $W_t^* := W_t - \lambda t$ is $\mathbb{P}^*$ -Brownian Motion. Then

$$f(X)/g(X) = e^{-\lambda \sum X_n + \frac{1}{2}\lambda^2 T} = e^{-\lambda W_T + \frac{1}{2}\lambda^2 T} = e^{-\lambda W_T^* - \frac{1}{2}\lambda^2 T}$$

Multiply the payoff $h(X)$ by this.

- ▶ When simulating under the measure $\mathbb{P}^*$, the zero-mean normals that you generate are simulations of W^* , not W .
- ▶ The $f(X)/g(X)$ is also known as the *Radon-Nikodym derivative* of $\mathbb{P}$ with respect to $\mathbb{P}^*$.

Example: Changing the drift of BM

Choice of λ

- ▶ How to choose λ ? Maybe too difficult to make S_T 's distribution agree entirely with the optimal distribution. So let's just make S_T 's *mean* $\approx$ the optimal distribution's *mean*. If S is GBM,

$$dS_t = rS_t dt + \sigma S_t dW_t = (r + \sigma\lambda)S_t dt + \sigma S_t dW_t^*$$

So $\mathbb{E}^* S_T = S_0 e^{(r+\sigma\lambda)T}$. You could, for instance, try to choose λ such that $S_0 e^{(r+\sigma\lambda)T} \approx$ the optimal distribution's mean.

- ▶ Even if λ is not chosen optimally, the importance sampling MC estimate is still unbiased.

Variance reduction: Antithetic variates

Variance reduction: Control variates

Variance reduction: Importance sampling

Variance reduction: Conditional MC

Conditional Monte Carlo

For any random variable X , by the law of iterated expectations,

$$C = \mathbb{E}Y = \mathbb{E}[\mathbb{E}(Y|X)] = \mathbb{E}f(X)$$

Assuming that the function $f(X) := \mathbb{E}(Y|X)$ can be computed analytically, Conditional Monte Carlo does the following:

- ▶ Simulate not Y but rather X , generating $X_1, \dots, X_M$ IID as X .
- ▶ Estimate $\mathbb{E}[\mathbb{E}(Y|X)]$ by taking the average

$$\hat{C}_M^{\text{cmc}} := \frac{1}{M} \sum_m f(X_m).$$

The standard error is $\hat{\sigma}_M^{\text{cmc}}/\sqrt{M}$ where

$$\hat{\sigma}_M^{\text{cmc}} := \sqrt{\frac{1}{M-1} \sum_m (f(X_m) - \hat{C}_M^{\text{cmc}})^2}$$

Conditional Monte Carlo: variance analysis

Expectation: The CMC estimate is unbiased, because

$$\mathbb{E}\hat{C}_M^{\text{cmc}} = \mathbb{E}f(X) = C.$$

Variance: CMC has smaller variance than ordinary MC because

$$\text{Var}(\hat{C}_M^{\text{cmc}}) = \frac{1}{M} \text{Var}f(X) = \frac{1}{M} \text{Var}[\mathbb{E}(Y|X)] \leq \frac{1}{M} \text{Var}(Y) = \text{Var}(\hat{C})$$

where the last step is because $\text{Var}(Y) = \text{Var}[\mathbb{E}(Y|X)] + \mathbb{E}\text{Var}(Y|X)$.

Intuition: suppose $Y = F(\mathbf{X})$ where $\mathbf{X} \in \mathbb{R}^D$. Then ordinary MC computes a D -dimensional integral $\int F(\mathbf{x})p(\mathbf{x})d\mathbf{x}$ by drawing random $\mathbf{X}_m \in \mathbb{R}^D$ with density p . But if, along d of those dimensions, the integration can be performed *analytically*, then we only need to draw random vectors $X_m \in \mathbb{R}^{D-d}$. This reduces the noise in the estimate.

Conditional Monte Carlo: example 1

Let $0 < t_1 < t_2 < t_3$. Consider at time 0 an Asian-style call on a GBM S , with expiry t_3 and discounted payoff

$$Y := e^{-rt_3} \left(\frac{S_{t_1} + S_{t_2} + S_{t_3}}{3} - k \right)^+$$

Let $X = (S_{t_1}, S_{t_2})$. Then $\mathbb{E}(Y|X)$ is analytically known, because

$$Y = \frac{1}{3} e^{-rt_3} \left(S_{t_3} - (3k - S_{t_1} - S_{t_2}) \right)^+$$

So $\mathbb{E}(Y|X)$ can be analytically evaluated using the B-S formula.

CMC: Generate M realizations of (S_{t_1}, S_{t_2}) and take the average of

$$\mathbb{E}(Y|X) = \frac{1}{3} e^{-rt_2} C^{BS}(S = S_{t_2}, t = t_2, K = 3k - (S_{t_1} + S_{t_2}), T = t_3).$$

This random variable has *smaller variance* than Y .

Conditional Monte Carlo: example 2a

Consider a call with discounted payoff $Y = e^{-rT}(S_T - K)^+$ where

$$dS_t = rS_t dt + \sqrt{V_t} S_t dW_{1t}$$

$$dV_t = \alpha(V_t) dt + \beta(V_t) dW_{2t}$$

and W_1 and W_2 are (in this particular example) *independent* BM.

Note that *conditional on the entire path of V_t* , the dynamics of S are

$dS_t = rS_t dt + \sqrt{V_t} S_t dW_{1t}$ where the V_t path is conditionally *fixed*,

and W_1 is still conditionally a BM. So Y has conditional expectation

$$\mathbb{E}(Y|X) = C^{BS}(\bar{\sigma}) \quad \text{where } \bar{\sigma}^2 := \frac{1}{T} \int_0^T V_t dt$$

and so Y has unconditional expectation given by the “mixing formula”

$$C = \mathbb{E} C^{BS}(\bar{\sigma}).$$

Conditional Monte Carlo: example 2a

Two approaches:

► Ordinary MC:

Generate M realizations of the two-dimensional path of (S, V) .

On each path calculate the discounted call payoff. Average.

► Conditional MC:

Generate M realizations of the one-dimensional path of V .

On each path calculate $C^{BS}(\bar{\sigma})$. Average.

CMC reduces variance by removing from the simulations the noise due to the randomness of S .

Conditional Monte Carlo: example 2b

Stochastic volatility model with nonzero correlation: Let

$$W_t = \sqrt{1 - \rho^2} dW_{1t} + \rho dW_{2t}$$

where W_1 and W_2 are independent BMs, so W and W_2 have correlation ρ . Let

$$dS_t = rS_t dt + \sigma_t S_t dW_t$$

$$\sigma_t = \sqrt{V_t}$$

$$dV_t = \alpha(V_t)dt + \beta(V_t)dW_{2t},$$

Conditional Monte Carlo: example 2b

Then $L := \log S$ satisfies

$$\begin{aligned} dL_t &= rdt - \frac{1}{2}\sigma_t^2 dt + \sigma_t \sqrt{1 - \rho^2} dW_{1t} + \sigma_t \rho dW_{2t} \\ &= rdt - \frac{1 - \rho^2}{2} \sigma_t^2 dt + \sigma_t \sqrt{1 - \rho^2} dW_{1t} - \frac{\rho^2}{2} \sigma_t^2 dt + \sigma_t \rho dW_{2t} \end{aligned}$$

So conditional on $\mathcal{F}_T^{W_2}$,

$$L_T \sim \text{Normal} \left(L_0 + rT + \log M_T(\rho) - \frac{1 - \rho^2}{2} \bar{\sigma}^2 T, (1 - \rho^2) \bar{\sigma}^2 T \right)$$

where

$$M_T(\rho) := \exp \left(-\frac{\rho^2}{2} \int_0^T \sigma_t^2 dt + \rho \int_0^T \sigma_t dW_{2t} \right)$$

So even with correlation, S is still W_2 -conditionally lognormal.

Conditional Monte Carlo: example 2b

“Mixing formula” for price of option paying $C_T = C(S_T)$

$$C_0 = \mathbb{E}(e^{-rT} C_T) = \mathbb{E}(\mathbb{E}(e^{-rT} C_T | \mathcal{F}_T^{W_2})) = \mathbb{E} C^{BS}(S_0 M_T(\rho), \bar{\sigma} \sqrt{1 - \rho^2})$$

where

$$M_T(\rho) := \exp \left(-\frac{\rho^2}{2} \int_0^T \sigma_t^2 dt + \rho \int_0^T \sigma_t dW_{2t} \right)$$

Option price is expectation of the **B-S formula for that option**,
evaluated at random volatility *and* a randomized spot.

Therefore:

Simulate W_2 . Given W_2 , evaluate $C^{BS}(S_0 M_T, \bar{\sigma} \sqrt{1 - \rho^2})$.

Average across all simulations of W_2 . No need to simulate W_1 .